

Q Rotate Image

T.C $\rightarrow O(n^2)$
S.C $\rightarrow O(n^2)$

Brute force

- ① $\rightarrow$ Create new matrix
- ② $\rightarrow$ Transpose the matrix
- ③ $\rightarrow$ Reverse every row
- ④ $\rightarrow$ return the matrix

Problem taking space complexity

Optimize

T.C $\rightarrow O(n^2)$
S.C $\rightarrow O(1)$

Code / Pseudo Code

```
// Transpose the matrix
for (int i = 0; i < row; i++)
    for (int j = i + 1; j < col; j++)
        swap(arr[i][j], arr[j][i]);
}
```

```
// Reverse single row :-
for (int i = 0; i < row; i++)
    s = 0, e = col - 1;
    while (s < e)
        swap(arr[i][s], arr[i][e]);
        s++; e--;
}
```

Dry Run

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

$\uparrow$ $\uparrow$
 0 1
 0 1

1	4	7
2	5	8
3	6	9
